Network Analysis on Rumination

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2. Methods

2.1 Network analysis

2.1.1 Removing redundant items

As a preparation step, we applied the *goldbricker* function from the R package *networktools* to search for pairs of items in the RRS, SCRS and MHS items which were highly correlated (r > 0.50) and showed highly the same pattern with other items. When such "bad pairs" were identified, reduction techniques were put into use (with the *net_reduce* function in R): instead of the pair, the first principal component of the two variables were kept in the data as a new, merged variable.

The RRS items required no reduction steps as the highly correlating items did not extend their pattern to more nodes (their correlations to other nodes did not violate the criteria that > 75% of these correlations did not differ significantly). Among the SCRS scales the SCRS_8 & SCRS_1 scales required reductions steps using PCA. Among the MHC nodes MHC_3 & MHC_1 items were subject to reduction.

2.1.2 Network Analysis with LASSO

For the network analysis, the tuning parameter of *gamma* in the grahical LASSO alogrithm has been set to 0.5 value in accordance with Bernstein et al in order to achieve sparse graphs. The final networks were chosen by the lowest values of the *extended Bayesian Information Criterion* (EBIC).

2.1.3. Community detection

In order to get an understanding of community of nodes and to see if nodes can be grouped into neighbourhoods, we used the *igraph* R package. When detecting possible communities of nodes, the detection happened with the following parameters in the *spinglass algorithm*: TODO: try the walktrap algorithm from igraph!

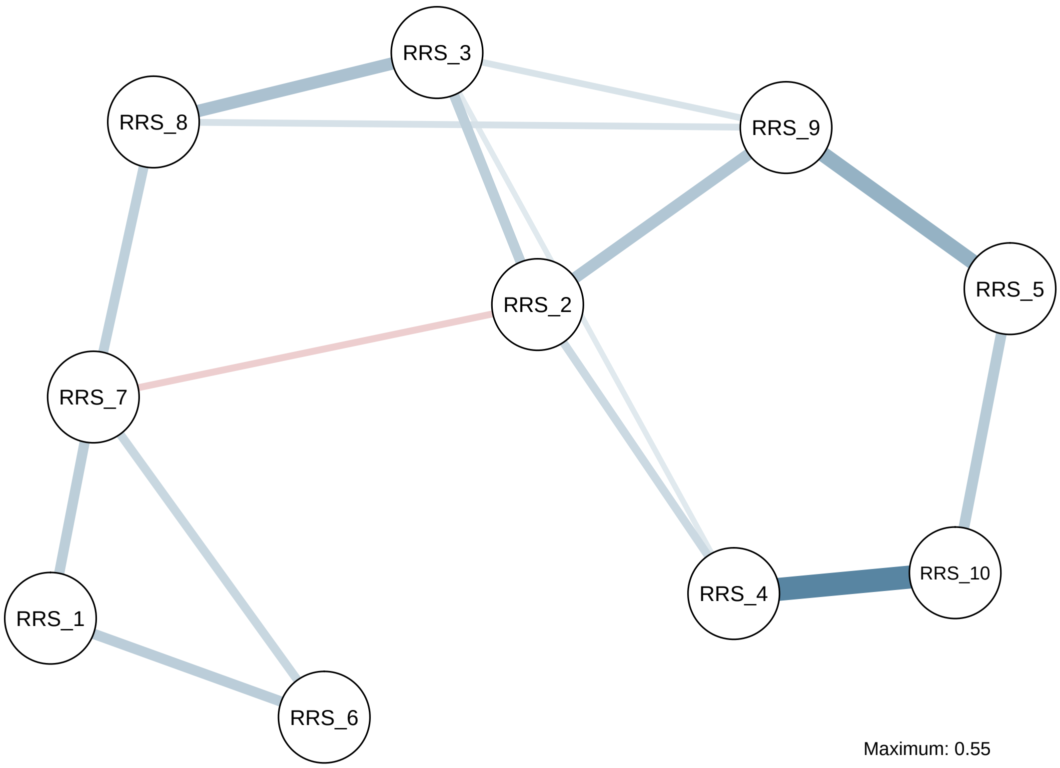
2.1.4. Expected Influence of nodes

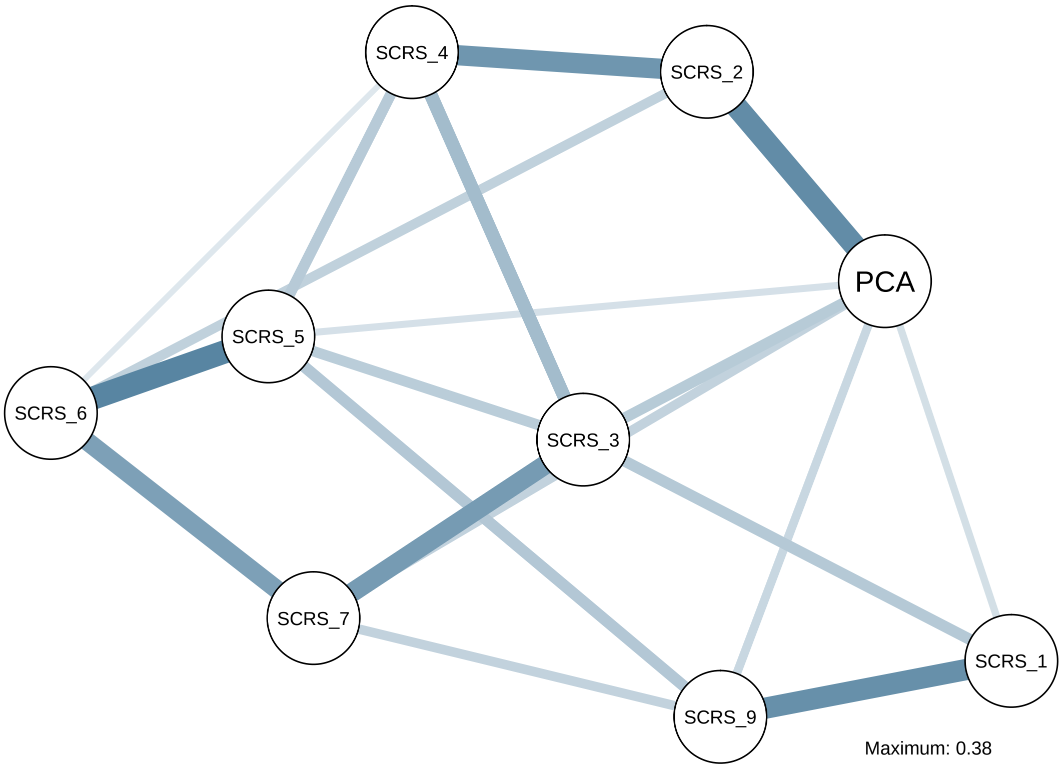
To understand certain items' centrality and importance, besides regularly used centrality measures (such as strength, closeness and betweenness of nodes in a network), one can also use the One- and Two-Step Expected Influence (EI) measures (Robinaugh et al, 2016). Traditionally in unweighted networks (where network edges are encoded to 0 or 1) a node's centrality is the sum of edges, and in weighted nodes it is the sum of the edge weights. In this approach negative edges are treated the same way as positives, as the absolute values are used for centrality estimation. On the other hand, the Expected Influence metrics take into account if a node has positive and negative edges, too.

3. Results

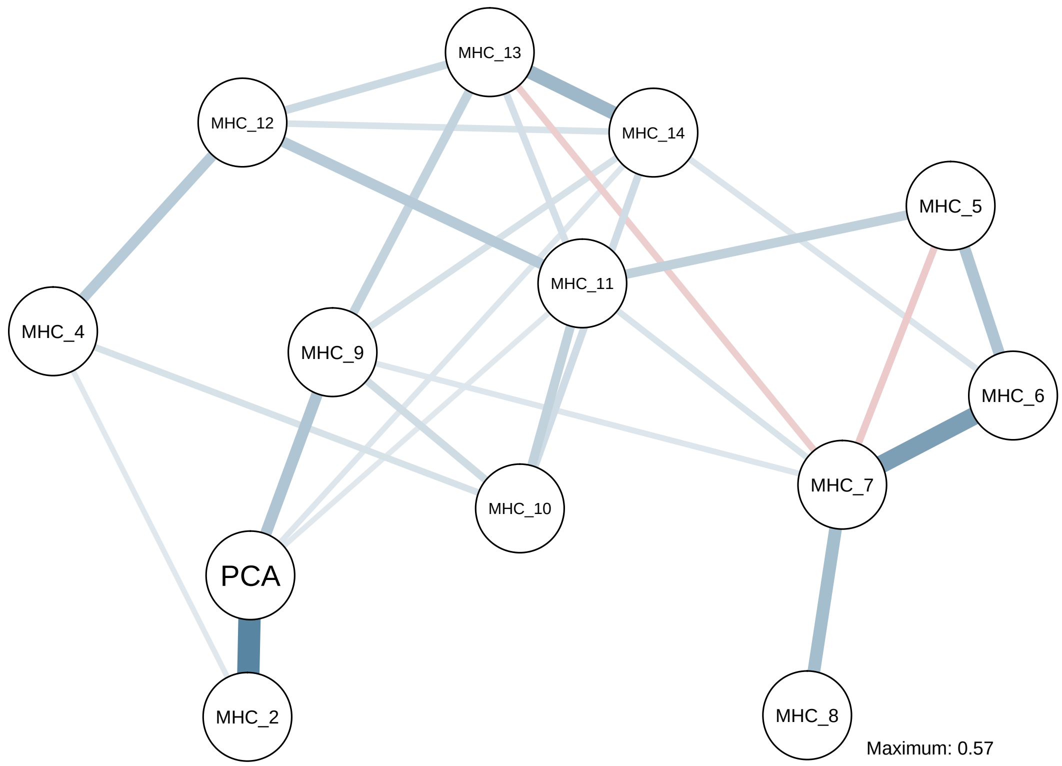
3.1 Network plots

RSS network





SCRS network

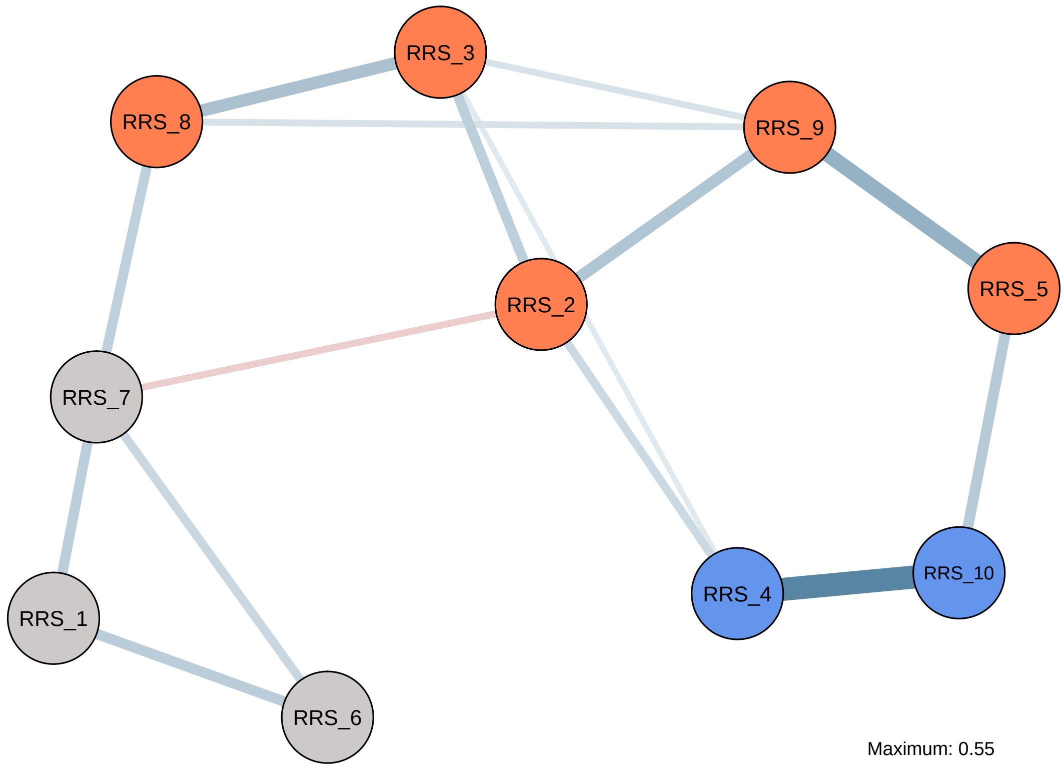


MHC network

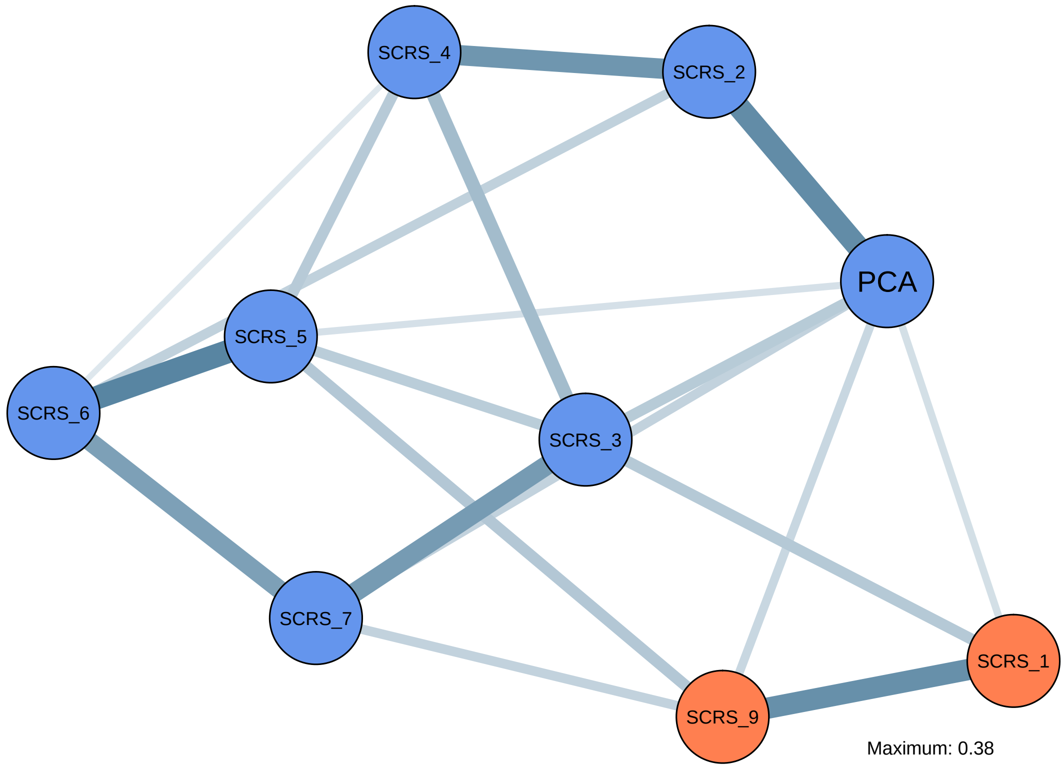
3.2 Community detection

Community mapping of RSS, MHC and SCRS with the *spinglass algorithm*:

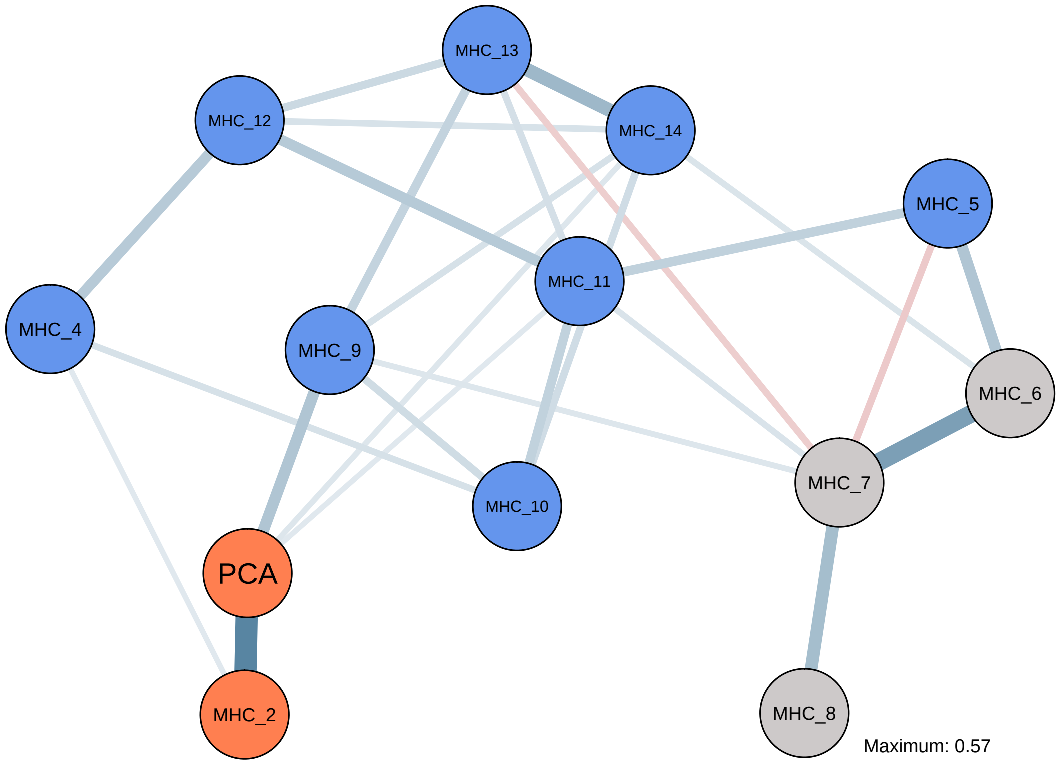
RSS communities identified with parameters *gamma = 0.5, spins = 25, starting temperature = 1, ending temperature = 0.01, cooling factor = 0.99*.



SCRS communities identified with parameters *gamma = 0.5, spins = 25, starting temperature = 1, ending temperature = 0.01, cooling factor = 0.99.*



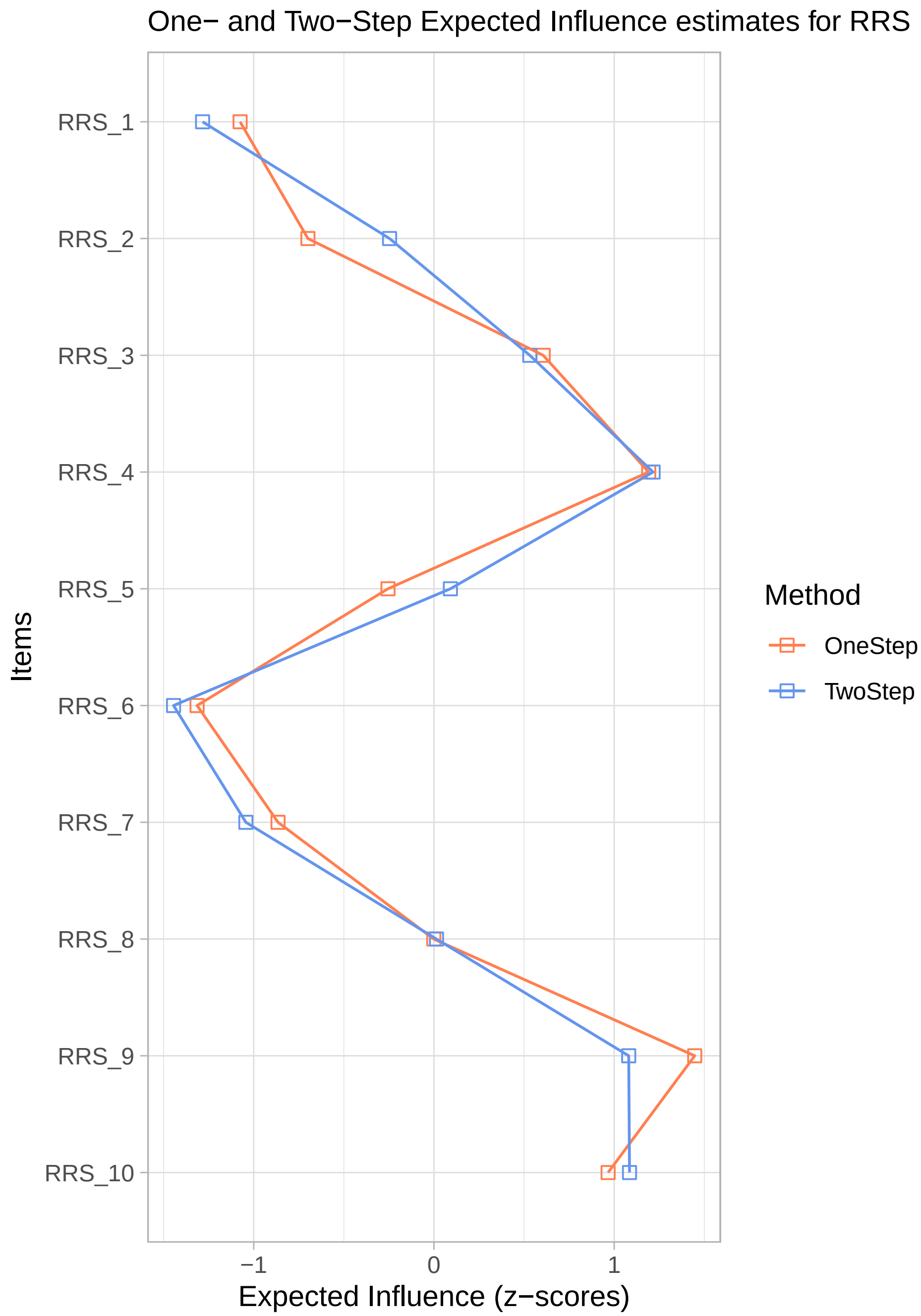
MHC communities identified with parameters *gamma = 0.5, spins = 25, starting temperature = 1, ending temperature = 0.01, cooling factor = 0.99.*



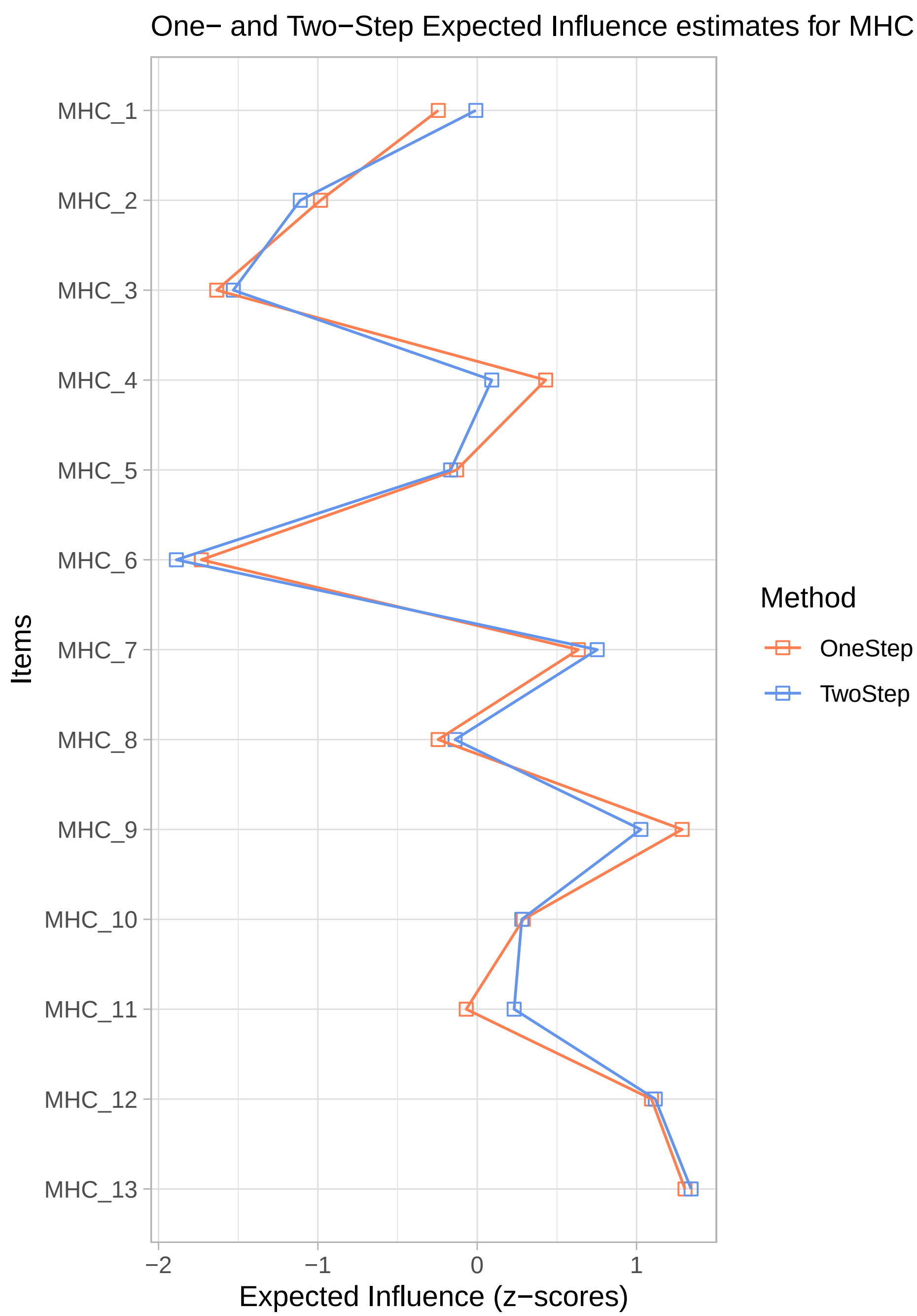
3.3 Expected Influence

In order to estimate the Expected Influence metrics, we used the *expectedInf* function of the *networktools* R package. The influence score are normalized.

RRS Expected Influence Measures



MHC Expected Influence Measures



SCRS Expected Influence Measures

References

Robinaugh, D. J., Millner, A. J., & McNally, R. J. (2016). Identifying highly influential nodes in the complicated grief network. Journal of Abnormal Psychology, 125. 747–757.